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Enhancing generalization in seismic image segmentation through multi-dataset integration and transfer learning

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Salt body segmentation in seismic images is vital for accurate hydrocarbon exploration. Deep learning models, specifically the U-Net, have shown a significant improvement for this task. However, existing models often lack generalizability when applied to diverse datasets, limiting their practical effectiveness. This limitation may arise due to factors such as data heterogeneity, variations in data preprocessing methods, differences in seismic acquisition parameters and geological settings, or even the choice of machine learning models, which may overfit to specific data characteristics and fail to adapt to unseen scenarios. This study addresses this gap by proposing a robust and generalizable segmentation framework trained on a combination of three diverse and widely used datasets – TGS, SEAM, and F3. The core of the methodology leverages an encoder–decoder architecture, where a pre-trained EfficientNet-B7 model, originally trained on ImageNet, serves as the encoder to significantly enhance feature extraction from complex seismic data. This integration of transfer learning with domain-specific training aims to improve model robustness and adaptability across varied geological settings. The model is evaluated independently on each dataset's test set, showing significant improvements in Intersection over Union (IoU), accuracy, and F1 score. Specifically, it achieves an IoU of 0.955 and an F1 score of 0.969 on the validation data, demonstrating strong segmentation capability and consistent generalization across varied geological conditions. These findings highlight the potential of multi-dataset training combined with transfer learning to build generalized salt body segmentation models suitable for real-world seismic applications.

Keywords. Salt; seismic; segmentation; deep learning; transfer learning.

1. Introduction

Accurate segmentation of salt bodies in seismic images is critical for hydrocarbon exploration, primarily because these structures often serve as traps for oil and gas reservoirs. Salt formations are complex in geometry, exhibit high acoustic impedance contrasts, and interfere with seismic wave propagation, making their interpretation particularly challenging. The task becomes even more complicated in deep-water regions where the subsurface environment is more intricate, directly affecting drilling decisions and risk assessments (Sheriff and Geldart 1995).

While traditional manual interpretation methods remain prevalent, they are labor-intensive, subjective, and impractical for large-scale surveys. Automated techniques using

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classical image processing or simple thresholding, fail to generalize due to the diverse geological settings and seismic acquisition parameters. Recent works have emphasized the importance of integrating geological context and seismic attribute analysis for improved subsurface interpretation (Narayan *et al.* 2024a, b). A significant limitation observed in many machine learning models is their overfitting to specific datasets, thereby reducing their capability to perform well on unseen data – a core generalization issue (Islam and Wali 2024).

Deep learning approaches, particularly U-Net and other encoder–decoders, have emerged as powerful tools in seismic interpretation. Studies have demonstrated the U-Net's potential for capturing complex salt body geometries (Shi *et al.* 2019; ul Islam 2020). However, despite advancements, many models are dataset-dependent and fail when applied to different geological basins or data collected under varying conditions. This poor generalizability is largely attributed to differences in noise levels, imaging artifacts, and salt morphology between datasets, which introduce significant variability (Muller *et al.* 2024). Additional factors such as data heterogeneity, variations in preprocessing techniques, differences in seismic acquisition parameters, and geological settings further exacerbate this challenge (Islam and Wali 2024). Moreover, the choice of machine learning models plays a critical role, as models overly tuned to specific dataset characteristics may overfit and struggle to adapt to unseen data, ultimately hindering their robustness and practical deployment in diverse exploration scenarios.

Recent advancements in seismic salt body segmentation have focused on improving the generalization and robustness of deep learning models across diverse geological scenarios. One notable approach is transfer learning, where pre-trained deep convolutional neural networks (CNNs) are fine-tuned for seismic tasks. Karchevskiy *et al.* (2018) demonstrated that applying transfer learning significantly enhances segmentation accuracy and model adaptability, particularly when applied to new geological settings. This method allows models to leverage knowledge from large generic datasets, reducing the need for extensive domain-specific labeling while improving generalization performance.

Semi-supervised learning has also emerged as a promising solution to the challenge of limited labeled seismic data. Babakhin *et al.* (2019) proposed an ensemble-based semi-supervised learning framework that effectively combines labeled and unlabeled data for salt body segmentation. This approach mitigates the heavy dependence on annotated datasets while improving the model's ability to adapt to complex and varied seismic features. By leveraging both labeled examples and the broader context provided by unlabeled data, the model achieves better performance on unseen samples.

Comprehensive reviews, such as the one by Islam and Wali (2024), have synthesized these advancements and identified persistent bottlenecks in generalization. They stressed the importance of training on diverse datasets to build models that are resilient to varying geological and acquisition conditions. Their analysis underscores the necessity for a broader dataset diversity to overcome overfitting and enhance model reliability in real-world seismic interpretation.

Further innovation comes from multitasking and 3D learning strategies designed to tackle the complexity of three-dimensional seismic data. Xu *et al.* (2025) introduced the 3D Saltseg-CL framework, which incorporates multi-task dense prediction and unsupervised learning components. This model enhances performance on 3D seismic volumes, improving the segmentation of complex salt structures by jointly learning multiple related tasks, thereby enriching the feature representation.

Robust data augmentation and regularization techniques have been explored to address overfitting issues in deep learning models. Zhang *et al.* (2023) emphasized the use of structural dropout layers and extensive data augmentation to improve the robustness of segmentation models. These techniques ensure the model can generalize better to new, unseen seismic datasets, reducing the risk of model degradation when deployed in real exploration scenarios.

Recent studies have proposed several innovative approaches to improve the robustness and generalization of salt body segmentation models, including transfer learning, semi-supervised learning, 3D multi-task frameworks, and advanced regularization techniques. While these methods show significant improvements in handling complex salt geometries and limited labeled data, a critical limitation remains – most of the models have been tested on specific datasets with limited geological diversity. The generalizability of these models across varied seismic datasets and real-world geological settings

has not been thoroughly validated, leaving a gap in confirming their robustness and adaptability in diverse exploration environments.

Building upon these insights, our proposed strategy integrates three diverse datasets – TGS, SEAM, and F3 – to expose the model to varying geological contexts, imaging conditions, and noise artifacts. We also utilized a transfer learning approach to train a more robust and generalizable model capable of handling diverse seismic datasets. Specifically, the framework leverages a pre-trained EfficientNet-B7 encoder, originally trained on the large-scale ImageNet dataset, to enhance feature extraction from complex seismic images. This powerful encoder is coupled with two decoder architectures – U-Net and LinkNet, which are well-known for their effectiveness in semantic segmentation tasks. The combination of EfficientNet-B7's rich feature representation and the spatial reconstruction capabilities of U-Net and LinkNet enables the model to capture intricate salt body geometries while preserving fine structural details. By integrating this transfer learning strategy with diverse datasets, we aim to improve generalization and adaptability, addressing one of the industry's most pressing challenges in seismic interpretation – developing models that perform reliably across varying geological settings and seismic acquisition conditions.

Some of the main contributions of this study are given below:

- Proposed a generalized salt body segmentation model by combining three diverse seismic datasets (TGS, SEAM, F3).
- Utilized a pre-trained encoder in an encoder–decoder architecture to enhance feature extraction and model robustness.
- Achieved superior segmentation performance across independent test sets, showcasing improved generalizability.

2. Methodology

The methodology section outlines the systematic approach employed in this research for salt dome segmentation in seismic images. It is divided into two key subsections: Data retrieval and preprocessing, which describe the acquisition and preparation of the datasets to ensure quality and consistency, and data modeling, which provides an overview of the encoder–decoder architectures utilized, along with details of model training and evaluation. Basic workflow is illustrated in figure 1.

2.1 *Data retrieval and preprocessing*

2.1.1 *Dataset*

The data preprocessing phase involved the integration of three publicly available datasets: TGS, SEAM, and F3, to create a unified dataset for training and evaluation. The TGS dataset, derived from the TGS Salt Identification Challenge (TGS 2024), comprises 101 × 101 grayscale seismic images with corresponding binary salt masks, along with a depth file providing depth information for each image. Its training set includes 4000 images with masks, while the test set contains images without masks.

The SEAM dataset (SEAM 2024) features three labeled seismic slices (inline numbers 4403, 4499, and 4595), with pixel-level salt and non-salt classifications. These seismic slices were divided into image patches, each with a corresponding mask. Lastly, the F3 dataset (F3 2024) consists of four labeled inlines (190, 290, 390, and 490) used for seismic structure segmentation. Originally comprising 13 seismic structures, its labels were simplified into a binary classification indicating the presence or absence of salt bodies. Figure 2 illustrates the class distribution of salt and non-salt pixels across three datasets.

2.1.2 *Data preprocessing*

The preprocessing pipeline involved several steps to prepare the dataset for model training and evaluation. Images and masks were first loaded

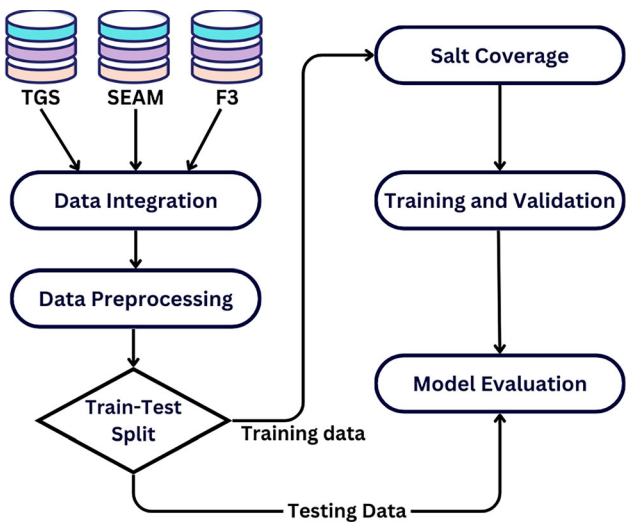


Figure 1. Methodology diagram.

into a Pandas DataFrame, and pixel values were normalized by dividing them by 255. Salt coverage for each mask was calculated as the ratio of salt pixels to total pixels, which was later used for stratification.

The data was split into training and testing sets with an 80–20 ratio. For testing, inline number 4403 from the SEAM dataset and 190 from the F3 dataset were reserved, while the rest were allocated to the training set. The TGS dataset, already in patch format, also had 20% of its data reserved for testing. A stratified train-validation split was then performed on the training data, ensuring that the salt coverage distribution in the training and validation sets closely matched the overall dataset distribution.

All images were resized to 128 × 128 × 3 dimensions to align with the input requirements of various encoder architectures. Data augmentation was applied by horizontally flipping both training images and their corresponding masks, effectively doubling the size of the training dataset. This augmentation step improved model generalization by accounting for the varying orientations of salt domes in seismic data.

2.2 *Model architecture*

This study employs an encoder–decoder architecture for binary image segmentation, leveraging a pre-trained EfficientNetB7 encoder known for its strong performance in feature extraction (Tan and Le 2019). Two decoder architectures were explored: U-Net (Ronneberger *et al.* 2015) and LinkNet (Chaurasia and Culurciello 2017), to compare their effectiveness in segmentation tasks. Figure 3 illustrates the encoder–decoder architecture.

The U-Net architecture (figure 4) features a symmetric U-shaped structure with extensive skip connections between encoder and decoder layers. These connections merge high-resolution feature maps from the encoder with upsampled outputs from the decoder, enabling precise localization and detail preservation, which are essential for segmentation. Conversely, LinkNet (figure 5), a lighter and more efficient architecture, employs shortcut connections both within its blocks and between the encoder and decoder stages. This design facilitates feature reuse, reduces computational complexity, and accelerates inference while maintaining competitive segmentation accuracy.

2.3 *Experimental setup*

The models were implemented using the EfficientNetB7 backbone with U-Net and LinkNet decoders from the segmentation_models library (Iakubovskii 2019). The segmentation_models library is a high-level deep learning framework built on top of Keras and TensorFlow for easy implementation of popular image segmentation architectures. It provides pre-trained encoder backbones (e.g., EfficientNet, ResNet) and modular decoder structures like U-Net, FPN, PSPNet,

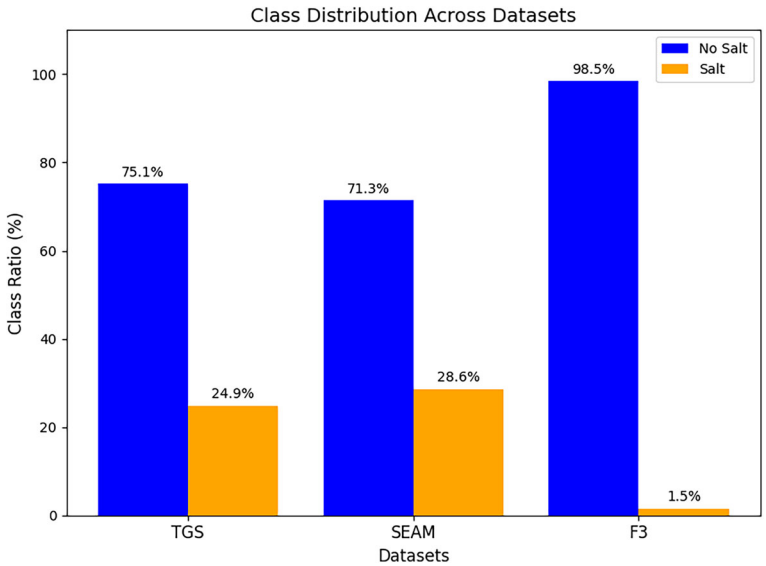


Figure 2. Bar chart illustrating the class distribution across TGS, SEAM, and F3 datasets, showing the percentage of salt and no-salt in each dataset.

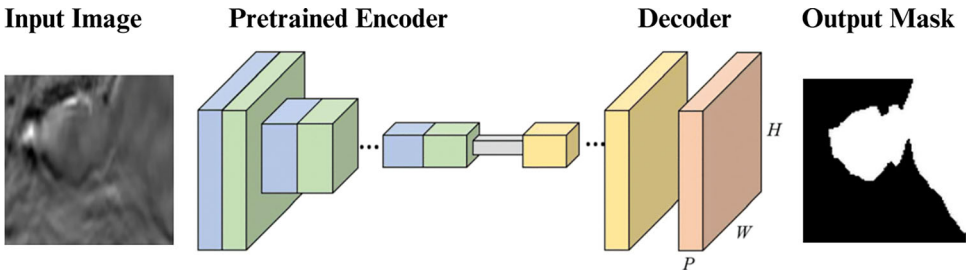


Figure 3. Pretrained encoder–decoder architecture.

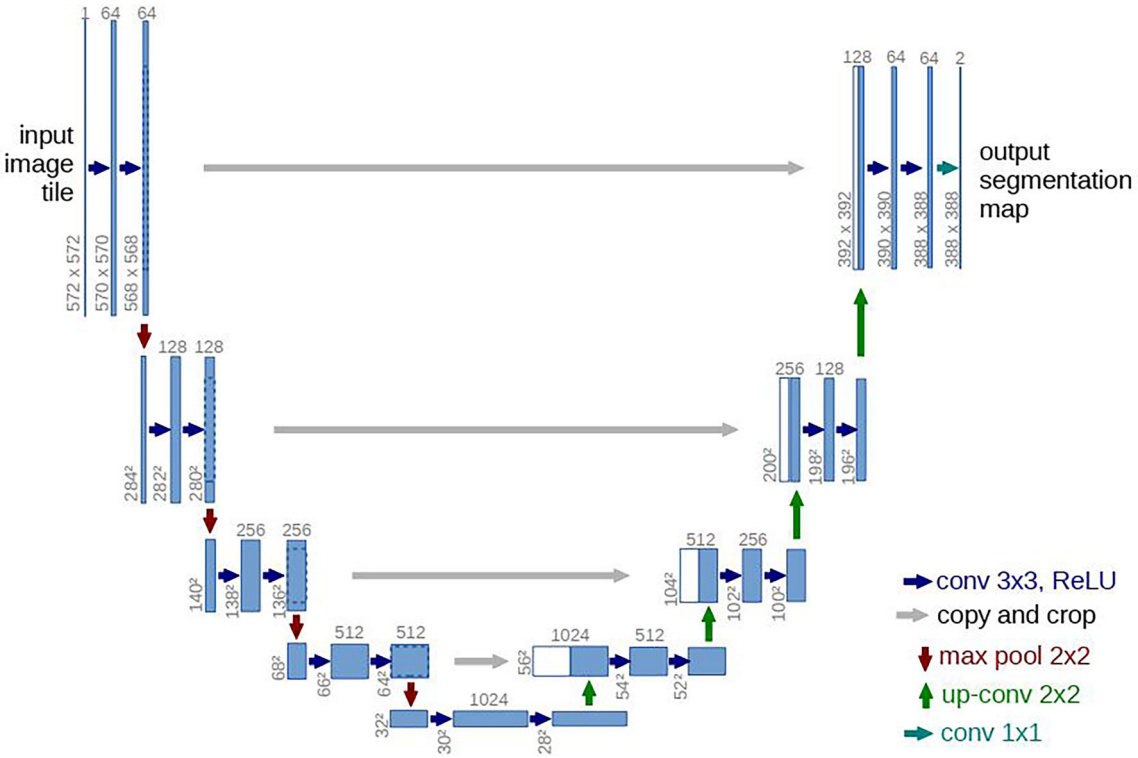


Figure 4. U-Net architecture (Ronneberger *et al.* 2015).

and LinkNet. Each model was trained for 100 epochs with a batch size of 32. Training utilized the Stochastic Gradient Descent (SGD) optimizer with momentum, Nesterov acceleration, and L2 regularization to control gradients and reduce overfitting.

To address class imbalance and enhance segmentation accuracy, a combined loss function was implemented, integrating Binary Cross-Entropy (BCE) with a smoothed version of Dice loss. The Dice loss is computed with a smoothing factor to ensure numerical stability, even when target regions are very small or absent. The total loss $\mathcal{L}_{\text{total}}$ is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}. \tag{1}$$

Additionally, Stochastic Weight Averaging (SWA) was applied during the final 30 epochs of training to stabilize convergence and improve generalization. Instead of storing all model weights from each epoch, SWA performs an online update to compute the averaged weights efficiently. Beginning from a specified epoch E_{SWA}, the model weights are updated iteratively using:

$$\bar{w}^{(t)} = \frac{1}{t - E_{\text{SWA}} + 1}\left(w^{(t)} + (t - E_{\text{SWA}})\,\bar{w}^{(t-1)}\right) \tag{2}$$

where $\bar{w}^{(t)}$ denotes the updated SWA weights at epoch t, $w^{(t)}$ are the model weights at that epoch, and E_{SWA} is the epoch at which SWA averaging

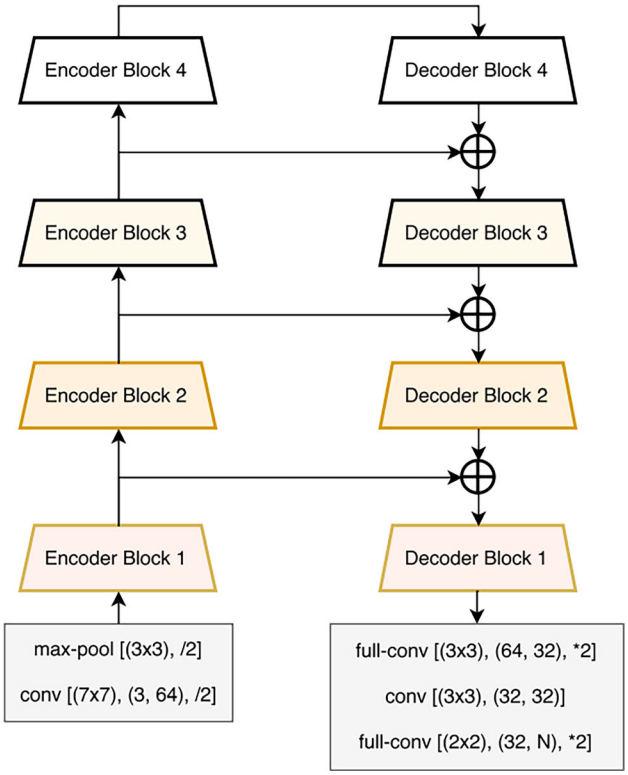


Figure 5. LinkNet architecture (Chaurasia and Culurciello 2017).

begins. At the end of training, the model is updated with these averaged weights and saved.

To promote stable convergence and prevent the optimizer from getting trapped in local minima, a cosine annealing learning rate schedule was employed. This scheduling technique gradually decreases the learning rate following a cosine function, allowing periodic restarts and encouraging the model to explore new minima during training. The learning rate at iteration t is computed as:

$$\alpha_t = \frac{\alpha_0}{2}\left(\cos\left(\frac{\pi \bmod (t, \frac{T}{M})}{\frac{T}{M}}\right) + 1\right) \quad (3)$$

where α_0 is the initial learning rate, T is the total number of epochs, M is the number of cosine cycles (snapshots), and t is the current epoch. This approach allows the learning rate to decay and restart cyclically, which can help escape sharp local minima and lead to better generalization.

2.4 *Model evaluation*

The trained models were evaluated on the validation and test sets using key metrics, including Intersection over Union (IoU), precision, recall, F1-score, and accuracy. IoU served as the primary evaluation metric, with a thresholding step applied to convert the model's continuous probability outputs into binary masks. To optimize IoU, thresholds ranging from 0.3 to 0.7 were tested in increments of 0.01, selecting the threshold that maximized IoU on the validation set.

Beyond IoU, additional metrics such as precision, recall, and F1-score were calculated to provide a balanced assessment of segmentation performance, highlighting the trade-off between over-segmentation (low precision) and under-segmentation (low recall). Accuracy was also computed to evaluate overall classification effectiveness. This comprehensive approach ensured a robust evaluation of the model's performance across diverse metrics, complementing the primary IoU analysis.

3. Results and discussion

In this section, we present the performance evaluation of the fine-tuned EfficientNetB7 encoder with U-Net and LinkNet decoders for binary image segmentation. The models were assessed on the validation set and three distinct test sets (TGS, SEAM, F3) using key metrics, including IoU, precision, recall, F1-score, and accuracy.

3.1 *Fine-tuning U-Net decoder*

The U-Net decoder paired with the EfficientNetB7 encoder demonstrated strong performance across the validation and test sets (table 1). On the validation set, it achieved an IoU of 0.955 and a high F1-score of 0.9694. Among the test sets, the SEAM dataset exhibited the best performance with an IoU of 0.932 and an F1-score of 0.9742, while the F3 dataset recorded the lowest IoU of 0.786 and an F1-score of 0.8366, reflecting challenges in segmenting this dataset.

3.2 *Fine-tuning LinkNet decoder*

The LinkNet decoder with EfficientNetB7 also exhibited robust performance, achieving similar results on the validation set with an IoU of 0.955 and an F1-score of 0.9693 (table 2). For the test sets, the highest performance was observed on the TGS dataset with an IoU of 0.880 and an F1-score of 0.9384, while the F3 dataset showed lower metrics, including an IoU of 0.790 and an F1-score of 0.7396, suggesting sensitivity to dataset variability.

Table 1. *Evaluation metrics for the U-Net–EfficientNetB7 model tested on the validation set and individual test sets (TGS, SEAM, F3).*

Dataset	IoU	Precision	Recall	F1-score	Accuracy
Validation	0.955	0.9832	0.9560	0.9694	0.9901
Testing (TGS)	0.879	0.9307	0.9338	0.9322	0.9662
Testing (SEAM)	0.932	0.9890	0.9598	0.9742	0.9858
Testing (F3)	0.786	0.8313	0.8420	0.8366	0.9792

Table 2. *Evaluation metrics for the LinkNet–EfficientNetB7 model tested on the validation set and individual test sets (TGS, SEAM, F3).*

Dataset	IoU	Precision	Recall	F1-score	Accuracy
Validation	0.955	0.9701	0.9684	0.9693	0.9899
Testing (TGS)	0.880	0.9460	0.9309	0.9384	0.9696
Testing (SEAM)	0.845	0.9854	0.8869	0.9335	0.9647
Testing (F3)	0.790	0.8618	0.6478	0.7396	0.9711

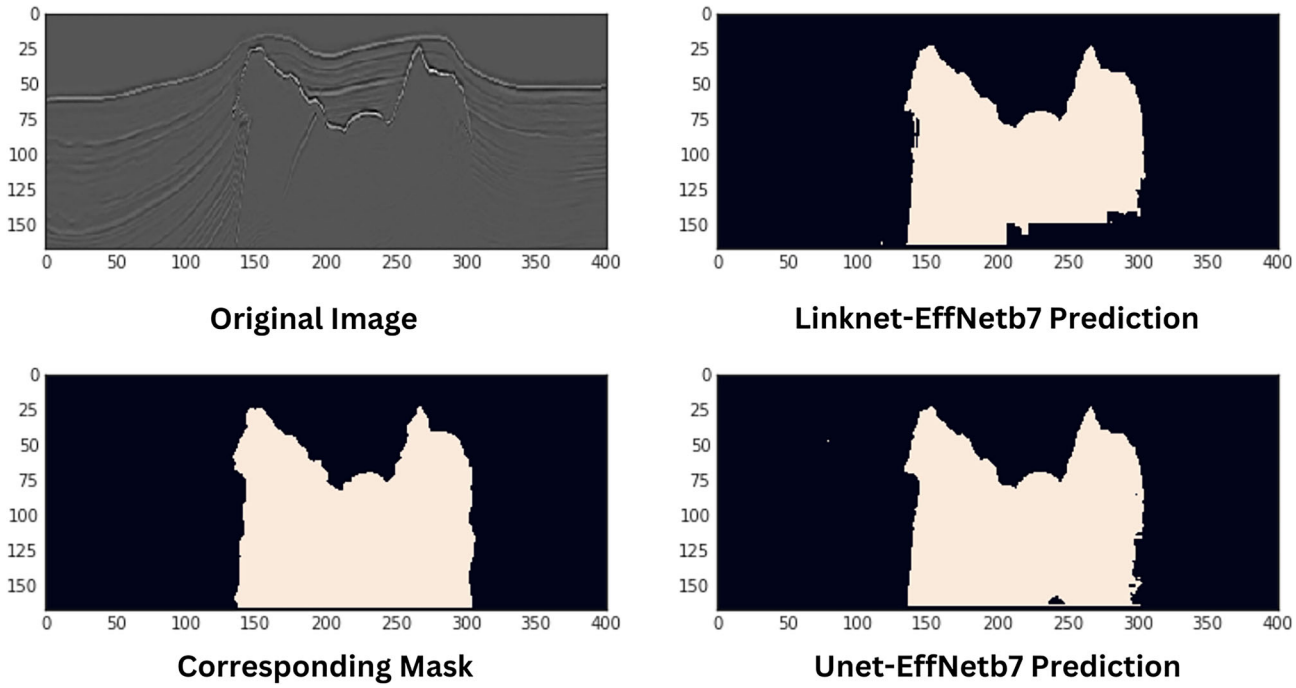


Figure 6. SEAM dataset: Original images with actual masks and predicted masks.

3.3 *Performance comparison of U-Net and LinkNet decoders*

Comparative analysis reveals that both decoders achieved comparable results on the validation set, but notable differences emerged across the test sets. The U-Net decoder outperformed LinkNet on the SEAM dataset with an IoU of 0.932 *vs.* 0.845 and on the F3 dataset with a smaller performance drop. However, LinkNet exhibited higher precision and slightly better results on the TGS dataset. These results underscore the strengths of U-Net in handling diverse datasets, while LinkNet demonstrated strengths in precision for specific scenarios.

Figures 6 and 7 illustrate the qualitative differences between the two decoders. In the SEAM dataset (figure 6), U-Net predictions showed better alignment with ground truth masks, particularly

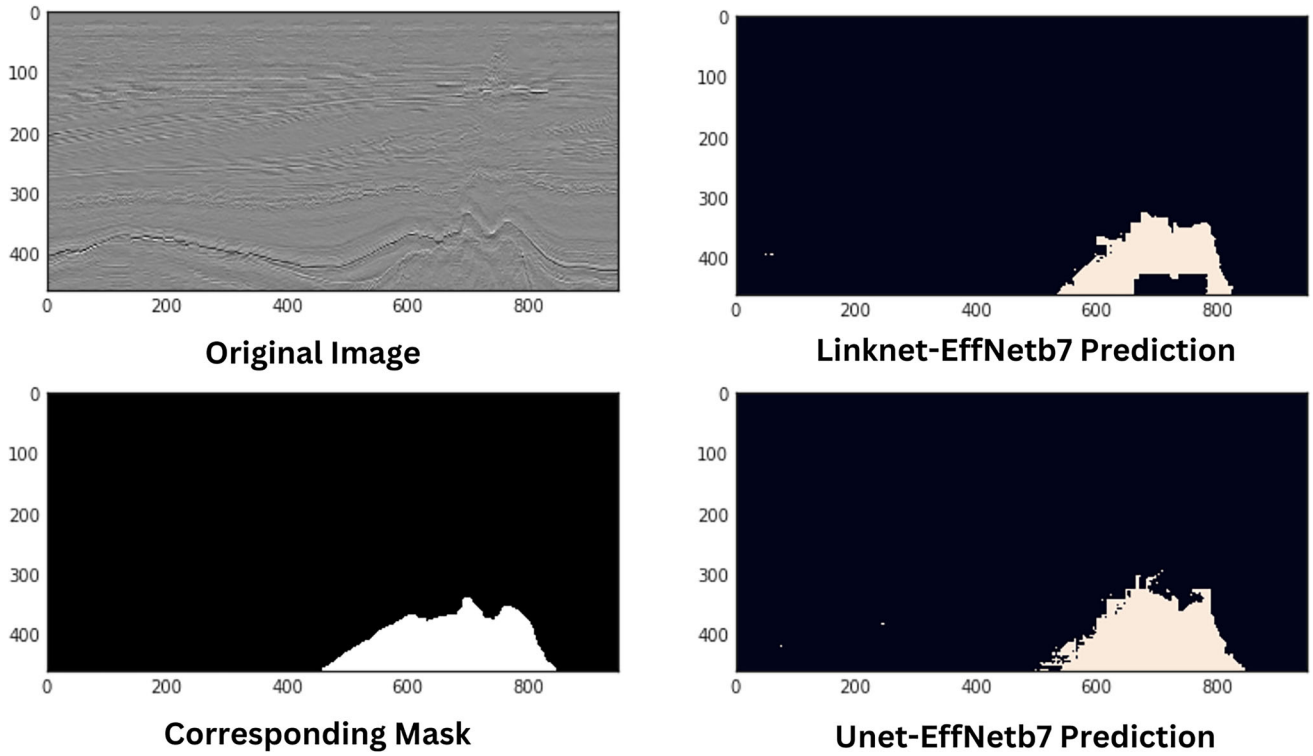


Figure 7. F3 dataset: Original images with actual masks and predicted masks.

Table 3. *Comparison of salt segmentation performance with related works on the TGS dataset. All models are evaluated using IoU, with accuracy and F1-score reported where available.*

Reference	Training	Model	Accuracy	F1-score	IoU
Typical baseline	Scratch	U-Net	92.0%	0.85	0.73
ul Islam (2020)	Scratch	U-Net + SE-ResNet	–	–	0.84
Al-Duri and Abbas (2021)	Scratch	U-Net	94.8%	–	–
Naoui *et al.* (2021)	Scratch	U-Net	–	–	0.70
Li *et al.* (2022)	TL	U-Net + ResNeSt-269	–	–	0.8732
Guo *et al.* (2020)	TL	U-Net + ResNet-34	–	–	0.8739
Cheng and Ma (2022)	TL	DeepLabV3 + ResNet-50	94.4%	–	0.857
Ours	TL	U-Net + EfficientNet-B7	**96.62%**	**0.932**	**0.879**

for finer structural details, reflecting its higher IoU and F1-scores. Conversely, in the F3 dataset (figure 7), both decoders struggled to segment complex regions, but LinkNet predictions displayed more fragmented results, aligning with its lower recall and F1-score. These visual comparisons complement the quantitative analysis, highlighting the U-Net's robustness in challenging scenarios.

3.4 *Class imbalance in F3 dataset*

The F3 dataset posed significant challenges for both models, primarily due to severe class imbalance, as indicated by the class distribution where salt regions constitute only 1.5% of the dataset. This imbalance likely contributed to the model's lower recall and F1-score, as the model struggled to detect and segment the minority class effectively. To address this, we incorporated a combined Binary Cross-Entropy and Dice loss, which improves sensitivity to underrepresented regions and stabilizes learning. Despite achieving relatively high precision, the difficulty in capturing salt regions hindered the overall segmentation performance, leading to a lower IoU compared to the other datasets. These results underscore the need for specialized techniques to address class

imbalance, especially in datasets like F3, where minority class representation is minimal.

3.5 *Comparison with previous work*

Table 3 presents a comparative analysis of the proposed method against several existing approaches for salt body segmentation. Models trained from scratch, such as standard U-Net or variants like U-Net with SE-ResNet, have demonstrated moderate performance, with IoU values generally ranging from 0.70 to 0.84. Transfer learning (TL) methods, including U-Net with ResNeSt-269 and DeepLabV3+ with ResNet-50, show improved IoU scores (above 0.85), highlighting the benefits of pre-trained encoders. Our method, which integrates EfficientNet-B7 as the encoder within a U-Net framework, outperforms prior approaches across all metrics, achieving an accuracy of 96.62%, an F1-score of 0.932, and an IoU of 0.879. This demonstrates the effectiveness of combining transfer learning with multi-dataset training for improving generalization in salt body segmentation.

Leveraging transfer learning with a powerful encoder like EfficientNet-B7 significantly enhances feature extraction and improves segmentation performance, particularly in complex seismic data. Although EfficientNet-B7 is computationally intensive, its inference time remains practical for offline or batch seismic processing, averaging around 20–30 ms per sample on an NVIDIA V100 GPU.

4. Conclusion

Accurate salt body segmentation remains a critical challenge in seismic interpretation, directly impacting hydrocarbon exploration and reservoir characterization. A significant obstacle is the poor generalizability of existing deep learning models when applied across diverse datasets, primarily due to variations in geological settings, seismic acquisition parameters, and data characteristics.

This study addressed this challenge by developing a robust and generalizable segmentation framework that leverages multi-dataset training using the TGS, SEAM, and F3 datasets. By integrating a transfer learning strategy – utilizing a pre-trained EfficientNet-B7 encoder with U-Net and LinkNet decoders – the model demonstrated superior feature extraction capabilities and improved segmentation performance. Comparative analysis revealed that while both decoders achieved strong validation results, the U-Net decoder excelled on the SEAM dataset, achieving an IoU of 0.932, and maintained better performance on the challenging F3 dataset. LinkNet, however, showed higher precision and slightly better results on the TGS dataset, highlighting the trade-offs between decoder choices.

The findings underscore the effectiveness of combining transfer learning with diverse multi-dataset training to enhance model robustness and generalization. This approach significantly improves key performance metrics – Intersection over Union (IoU), accuracy, and F1-score – across independent test sets, reinforcing the model's potential for practical deployment in real-world seismic interpretation tasks.

Despite these promising results, the study also highlights persistent limitations, particularly the impact of severe class imbalance in datasets like F3, which affects segmentation performance. Addressing class imbalance remains a critical area for future research, involving the incorporation of tailored data augmentation strategies, the exploration of specialized loss functions, and the investigation of advanced architectures or domain adaptation techniques to improve model resilience.

Author statement

Muhammad Saif ul Islam: Conceptualization, experimentation, writing – original draft. Aamir Wali: Supervision, writing – original draft.

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